1 Cosmology: DES Y1 3×2pt

Scope note. This is the PICKLES-focused note: the headline comparison is Metropolized PICKLES against NUTS (`NumPyro`) and `emcee`. All gradient-sampler results use the corrected step-size dual-averaging (Nesterov gain $\sqrt{m}/\gamma$); see `results/des_y1_fda/` for the directory-to-table map.

We benchmark on the Dark Energy Survey Year 1 (DES Y1) 3×2pt cosmological inference, implemented through the `JAX-COSMO` forward model. The posterior is $D = 21$ dimensional, with the standard cosmology reparametrization (logit transform on the 13 Uniform priors and non-centered location-scale on the 8 Gaussian priors), so the prior is $\mathcal{N}(0, I_{21})$ in the sampled z-space. Empirically the posterior has condition number $\kappa(\Sigma) \approx 300$, with six parameter pairs at $|\rho| > 0.9$ — all between σ_8 and the five galaxy bias parameters $b_{1\text{–}5}$ — and eight prior-dominated nuisance directions (Figure 1). PICKLES uses the empirical covariance of the live walker ensemble as a preconditioner, recomputed every iteration, in place of a separately-adapted mass matrix; its two tuning knobs are the kinetic-damping coefficient γ and the trajectory-length cap $L_{\max}$.

All runs use float64 on a single NVIDIA H100 GPU. ESS denotes the minimum effective sample size across the 21 dimensions, from the uncapped integrated autocorrelation time ($\hat{\tau} < 1$ allowed for antithetic chains); ESS / eval is the production-only count of target evaluations (gradients for HMC/PICKLES, log-probs for `emcee`). PICKLES rows are 3-seed medians at N=1000 (single seed at N=5000); NUTS uses deterministic `init_to_median`.

1.1 PICKLES is competitive with `emcee` and NUTS-dense

Table 1 reports worst-dimension ESS per gradient and wall-clock; all ESS values are computed in the common physical parameter space (see the parametrization note in §1.5). At the recommended tuning ($\gamma = 0.5$, $L_{\max} = 20$) PICKLES achieves worst-dimension ESS/grad = 5.11×10^{-3}, an effective tie with NUTS-dense (4.26×10^{-3}) from this fiducial-ball start — where one of three PICKLES seeds suffers a rare w_s=0 preconditioner collapse that widens the spread (§1.2, §1.5) — and above NUTS-block/diagonal ($3.93/2.29 \times 10^{-3}$), with zero numerical divergences. From a *matched, equilibrated* start (prior median plus a short gradient-free warm-start, the apples-to-apples comparison) the collapse disappears and PICKLES is a robust 1.44× ahead (6.14×10^{-3} vs 4.26, tight across seeds; Table 2). Per dimension, however, NUTS is marginally more gradient-efficient on the fast directions (§1.2); PICKLES's worst-dimension edge comes from a flatter ESS profile across coordinates, not from beating NUTS on any typical direction. The honest summary is that PICKLES *matches or slightly exceeds* dense-mass NUTS per gradient on this posterior, with its decisive gains lying elsewhere (warmup-robustness, divergence-freeness, parallelism). PICKLES is far ahead of gradient-free `emcee`, which needs $N \gg 1000$ to converge here; at matched long production both are converged and PICKLES leads by ∼24× (Table 3).

The damping coefficient γ is the dominant knob: lowering the default $\gamma = 2$ to $\gamma = 0.5$ improves ESS / grad by 1.4× (the empirical preconditioner already absorbs the scale anisotropy, so lighter damping helps). The trajectory cap $L_{\max}$, by contrast, has little effect on per-gradient efficiency at $\gamma = 0.5$ (5.1, 4.9, 5.1×10^{-3} for $L_{\max} = 20, 50, 128$): longer trajectories give proportionally more raw ESS at proportionally more cost. $L_{\max} = 20$ is therefore preferred only because it has the lowest wall-clock (∼ 3× faster than $L_{\max} = 128$ for the same per-gradient efficiency).

Fair, matched-initialization comparison. The main table starts PICKLES from a fiducial ball and NUTS from `init_to_median` (a ball at the prior median $z = 0$). To compare at an *identical* starting point, Table 2 reruns PICKLES from the same prior-median center NUTS uses (with a short gradient-free stretch pre-warmup so the ensemble reaches the typical set, the analogue of `init_to_median`). Recommended PICKLES then delivers worst-dimension ESS/grad = 6.14×10^{-3}, 1.44× ahead of NUTS-dense at the same init — the advantage is not an artifact of the initialization.

A long-production reference at $N_{\text{prod}} = 5000$ is given in Table 3; the ranking is preserved.

1.2 Per-parameter breakdown

Table 4 breaks the worst-dimension number down across all 21 parameters, revealing a *bimodal* regime shared by both samplers. Fifteen dimensions reach ESS / grad in the 1–3×10^{-2} range; the remaining six

Table 1: DES Y1 3×2pt sampler comparison at $N_{\rm prod} = 1000$ (3-seed median; walkers in a Gaussian ball of width 0.5 in z-space around the fiducial cosmology, no stretch pre-warmup; 500 gradient-sampler warmup). $\bar{L}$ is the mean trajectory length (adaptive for NUTS via the no-U-turn criterion). NUTS-block takes nearly twice as many leapfrog steps as NUTS-dense because its block-diagonal mass matrix cannot capture the σ_8–bias cross-correlations.

	$N_{\rm ch}$	wall (h)	$\bar{L}$	acc.	ESS	ESS / eval
Metropolized PICKLES (this work) — three values of $L_{\rm max}$:						
PICKLES ($\gamma = 0.5$, $L_{\rm max} = 20$, recommended)	64	1.5	20.0	0.92	6,540	$\mathbf{5.11 \times 10^{-3}}$
PICKLES ($\gamma = 0.5$, $L_{\rm max} = 50$)	64	3.2	48.5	0.88	15,248	4.92×10^{-3}
PICKLES ($\gamma = 0.5$, $L_{\rm max} = 128$)	64	4.3	63.5	0.87	20,605	5.07×10^{-3}
PICKLES (default $\gamma = 2$, $L_{\rm max} = 20$)	64	1.5	20.0	0.88	4,699	3.67×10^{-3}
PICKLES (default $\gamma = 2$, $L_{\rm max} = 50$)	64	3.2	48.6	0.88	4,783	1.55×10^{-3}
PICKLES (default $\gamma = 2$, $L_{\rm max} = 128$)	64	4.6	67.1	0.87	12,079	2.44×10^{-3}
Reference samplers:						
NUTS-dense (`numpyro`)	4	8.4	33	0.80	714	4.26×10^{-3}
NUTS-block (Campagne+ structure)	4	11.3	63	0.80	986	3.93×10^{-3}
NUTS-diag	4	12.5	83	0.80	727	2.29×10^{-3}
`emcee` / stretch (gradient-free)[♭]	64	0.05	—	0.27	96	1.50×10^{-3}

[♭] `emcee` has *not* converged at N=1000 ($\hat{R} = 1.8$): its long autocorrelation tail is unresolved, so this ESS / grad is only indicative. The converged comparison is the long run in Table 3.

Table 2: Matched-initialization comparison: all samplers started at the prior median ($z = 0$). PICKLES uses $w_s = 200$ gradient-free stretch pre-warmup (the ensemble analogue of `init_to_median`, zero gradients); NUTS uses `init_to_median`. $N_{\rm prod} = 1000$, 3-seed median.

sampler (prior-median init)	ESS	ESS / grad
PICKLES ($\gamma = 0.5$, $L_{\rm max} = 20$, $w_s = 200$)	7,863	$\mathbf{6.14 \times 10^{-3}}$
PICKLES ($\gamma = 2$, $L_{\rm max} = 20$, $w_s = 200$)	4,774	3.73×10^{-3}
NUTS-dense (`numpyro`)	714	4.26×10^{-3}

— σ_8 and the five galaxy biases — bottleneck at $\approx 6 \times 10^{-3}$, a 4–6× per-dimension penalty from mixing along the $|\rho| \approx 0.94$ ridge. The bottleneck is at the *same* dimensions and magnitude for both samplers, so it is structural, not sampler-specific. *Per gradient the two are close, but NUTS is marginally more efficient on most dimensions* (P/N < 1 on 16 of 21): NUTS's no-U-turn tree search produces near-i.i.d. draws very gradient-efficiently, whereas PICKLES accumulates $\sim 8\times$ more raw ESS while spending $\sim 10\times$ more gradients. PICKLES is level or slightly ahead only on the galaxy biases (ratios 0.88–1.18). On the bottleneck dimension that governs the benchmark the two are an *effective tie* (σ_8 at 5.11 vs b_1 at 5.08×10^{-3}). The small numerical edge PICKLES shows in Table 1 is *not* significant: that table reports the single worst dimension of each run (the minimum over 21 coordinates, the noisiest possible statistic), and with three seeds the worst-dimension ESS / grad is $[5.1, 7.1, 0.17] \times 10^{-3}$ for PICKLES versus $[7.6, 3.7, 4.3] \times 10^{-3}$ for NUTS — overlapping spreads whose medians (5.11 vs 4.26) differ within seed noise. The two have *different* failure modes rather than one being more consistent: NUTS's slow-subspace mixing varies seed-to-seed (no collapse), while PICKLES is steadier on σ_8 but carries a rare preconditioner collapse at w_s=0 (one of three seeds here; see §1.5). From the matched, equilibrated start of Table 2 the collapse is absent and PICKLES's worst-dimension ESS / grad is a robust 1.44× above NUTS — the worst-dimension metric rewards PICKLES's flatter cross-coordinate profile even though NUTS wins most *individual* dimensions. The fair reading: PICKLES *matches or slightly exceeds* dense-mass NUTS per gradient on this posterior; its decisive advantages are robustness to warmup, zero divergences, and 64-chain parallelism.

Table 3: DES Y1 3×2pt long-production reference at $N_{\rm prod} = 5000$ (single seed for all methods). `emcee` uses $N_{\rm prod} = 50{,}000$. Configuration as in Table 1.

	$N_{\rm ch}$	wall (h)	$\bar{L}$	acc.	div.	ESS	ESS / eval
PICKLES ($\gamma = 0.5$, $L_{\max} = 20$)	64	5.0	20.0	0.90	0	38,669	$\mathbf{6.04 \times 10^{-3}}$
PICKLES (default $\gamma = 2$)	64	4.7	20.0	0.90	0	20,727	3.24×10^{-3}
NUTS-dense (`numpyro`)	4	10.2	32	0.91	24	2,314	3.60×10^{-3}
`emcee` / stretch (gradient-free)	64	0.78	—	0.26	—	839	2.57×10^{-4}

1.3 Robustness to warmup length

Dense-mass NUTS adapts $D(D+1)/2 = 231$ mass-matrix entries by window adaptation during warmup, whereas PICKLES re-estimates its preconditioner from the live ensemble every iteration and only converges a scalar step size. We therefore expect PICKLES to tolerate a much shorter warmup. Table 5 and Figure 3 push the gradient-sampler warmup down to $w = 25$ steps — far below the $\gtrsim 100$ a 231-entry dense mass needs. The PICKLES warmup is 50 gradient-free stretch steps (to spread the ensemble to the typical set) followed by w ordinary steps (to converge the step size); NUTS uses w window-adaptation steps. Both start from the prior median, so the comparison is at an identical initialization.

PICKLES is dependable: flat and fully converged at every warmup, down to $w = 25$. Worst-dimension ESS / grad stays at 5–6×10^{-3}, $\hat{R}_{\max} = 1.01$, acceptance ≈ 0.92, fixed $L = 20$, and *zero* divergences (Fig. 3a, red). *NUTS-dense fails below $w \approx 100$ and sits under PICKLES at every warmup.* At $w = 25$ it *catastrophically* fails in *all three* seeds — $\hat{R}_{\max} = 1.13$, acceptance 0.4–0.7, 800–1400 divergences per seed, and worst-dimension ESS / grad of 8×10^{-5} (median), a factor ~ 70 below PICKLES at the same warmup; at $w = 50$ it still fails to converge ($\hat{R}_{\max} = 1.08$, 343 divergences). Only by $w \approx 100$ does it recover to PICKLES-comparable efficiency and convergence. The failure mode varies seed-to-seed — diverge or fail to converge — but the consequence is uniform: below $w \approx 100$ NUTS draws cannot be trusted without a convergence check, whereas PICKLES is reliable throughout.

The mechanism is that short-warmup dense-mass adaptation is itself unreliable: with too few samples the 231-entry estimate is noisy, and the resulting dynamics diverge or fail to converge rather than slow down gracefully. PICKLES has no separate mass-adaptation phase to get wrong — its preconditioner is the live ensemble covariance, recomputed every iteration — so it never enters this failure regime. This is the affine-invariance payoff for warmup robustness.

Table 4: Per-parameter ESS and ESS / grad at $N_{\text{prod}} = 1000$, 3-seed median (PICKLES $\gamma = 0.5$, $L_{\max} = 20$, fid-ball init, $w_s = 0$, $w = 500$; NUTS-dense `numpyro` defaults). Bold rows mark the 6-dim slow subspace.

	ESS		ESS / grad ($\times 10^{-3}$)		ratio
parameter	PICKLES	NUTS-d.	PICKLES	NUTS-d.	P/N
Cosmology					
Ω_c	26,288	3,111	20.54	23.32	0.88
$\boldsymbol{\sigma_8}$	**6,540**	**924**	**5.11**	**5.51**	0.93
Ω_b	34,599	3,837	27.03	28.77	0.94
h	20,725	2,639	16.19	20.01	0.81
n_s	30,126	3,834	23.54	28.74	0.82
w_0	17,171	1,540	13.41	12.31	1.09
Intrinsic alignment					
A	37,059	4,968	28.95	36.80	0.79
η	38,441	5,233	30.03	39.23	0.77
Galaxy bias (slow subspace)					
$\boldsymbol{b_1}$	**7,581**	**853**	**5.92**	**5.08**	1.17
$\boldsymbol{b_2}$	**7,839**	**829**	**6.12**	**5.19**	1.18
$\boldsymbol{b_3}$	**7,974**	**871**	**6.23**	**6.35**	0.98
$\boldsymbol{b_4}$	**6,920**	**889**	**5.41**	**6.15**	0.88
$\boldsymbol{b_5}$	**7,654**	**725**	**5.98**	**5.87**	1.02
Shear-calibration m_i					
m_1	39,382	4,930	30.77	36.96	0.83
m_2	39,126	4,664	30.57	34.97	0.87
m_3	40,418	4,594	31.58	33.87	0.93
m_4	40,759	5,184	31.84	38.34	0.83
Photo-z Δz_i					
Δz_1	40,453	5,675	31.60	42.55	0.74
Δz_2	40,114	4,994	31.34	37.44	0.84
Δz_3	40,026	5,174	31.27	34.14	0.92
Δz_4	40,668	4,532	31.77	30.13	1.05

Table 5: Robustness to gradient-sampler warmup length on DES Y1 3 × 2pt ($N_{\text{prod}} = 1000$; production-only worst-dimension ESS / grad, physical-space ESS, prior-median init). PICKLES is converged ($\hat{R}_{\max} = 1.01$) with 0 divergences and fixed $L = 20$ at every w; the NUTS $\hat{R}_{\max}$ and divergence rows are the warmup-health signals. All columns are 3-seed medians ($\hat{R}$ max, divergences summed over seeds).

ESS / grad ($\times 10^{-3}$)	w=25	w=50	w=100	w=200
PICKLES (γ=0.5, L=20, w_s=50), acc	**5.50** (.94)	**5.36** (.93)	**4.95** (.92)	**6.06** (.90)
NUTS-dense	0.081	1.27	4.04	1.93
NUTS warmup-health signals				
max split-$\hat{R}$	1.13	1.08	1.01	1.02
divergences	3126	343	5	32
mean trajectory $\bar{L}$	74	92	57	50

PICKLES holds $\hat{R}_{\max} = 1.01$, 0 divergences, and $L = 20$ at every w (omitted as constant). NUTS divergence counts at $w \geq 25$ are summed over 3 seeds; the $w = 25$ failure is uniform across seeds ($\hat{R} = 1.13/1.09/1.11$, 797/1382/947 divergences, acc 0.68/0.39/0.42), not a single unlucky run. The fragility is not cured by adding prior structure: block-mass NUTS, which exploits the survey's known parameter-block grouping (∼40 mass entries instead of 231), still produces 119 divergences at $w = 50$. PICKLES needs only the 50-step gradient-free stretch pre-warmup (no gradients, no target covariance, no structural prior) to be robust at every w; without it the worst-dimension ESS is seed-dependent (Table 10).

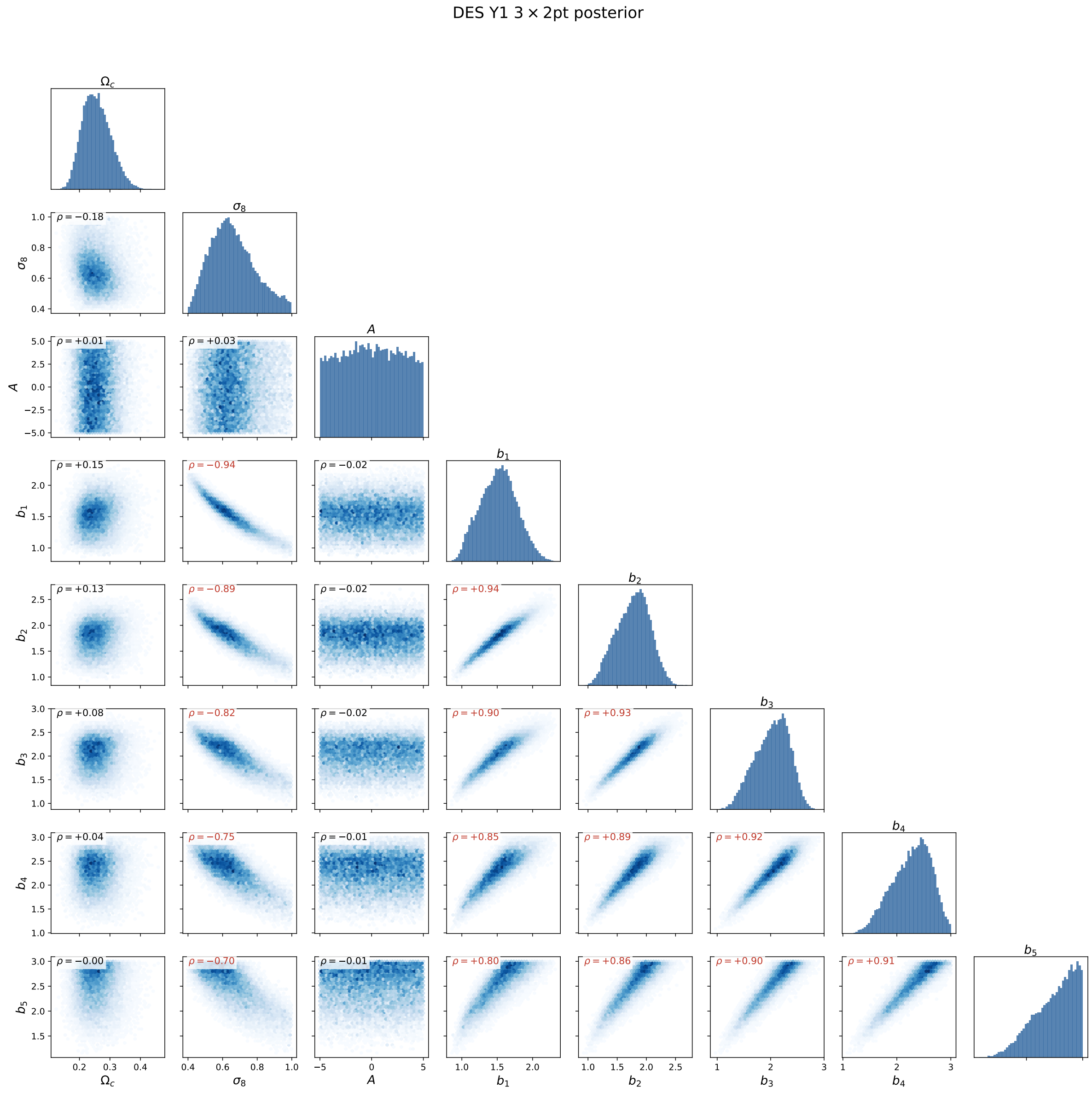

Figure 1: DES Y1 3 × 2pt posterior, eight *sampled* marginals of $D = 21$. The control panels (rows Ω_c, A) show near-isotropic ellipses with $|\rho| \le 0.2$, while every panel involving σ_8 and the five galaxy biases $b_{1:5}$ shows a narrow ridge ($|\rho|$ from 0.70 to 0.94). The slow-mixing 6-dim subspace consists entirely of $\sigma_8 + b_{1:5}$ (§1.2).

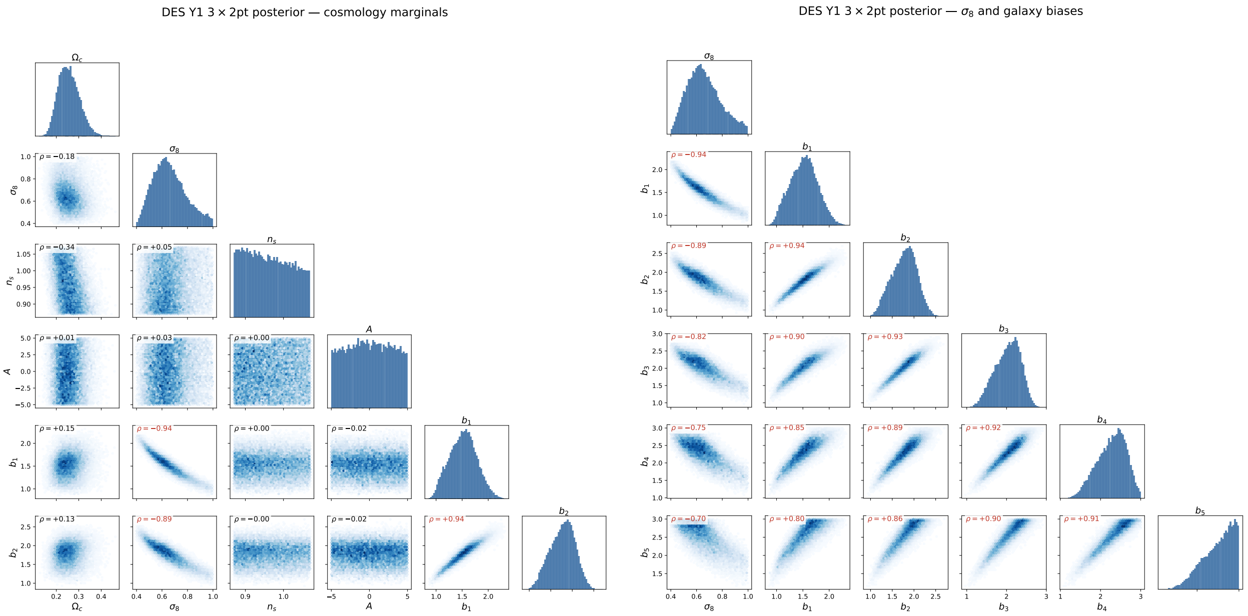

Figure 2: Alternative six-dimensional views. Left: the conventional cosmology corner. Right: the slow subspace ($\sigma_8 + b_{1:5}$), in which every off-diagonal panel is a narrow ridge ($|\rho|$ up to 0.94) — the 6-dim subspace responsible for the ESS bottleneck in Table 4.

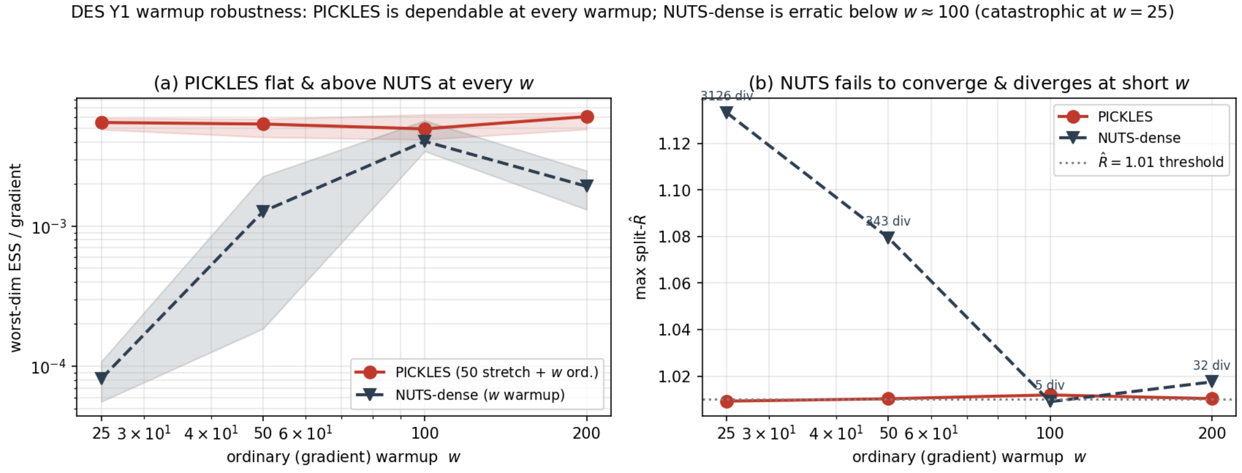

Figure 3: DES Y1 warmup robustness. **(a)** Worst-dimension ESS / grad (physical space) vs. gradient-sampler warmup: PICKLES (red, 50-step stretch $+w$ ordinary) is flat and above NUTS-dense (navy) at every warmup; the NUTS curve craters to $\sim 8 \times 10^{-5}$ (3-seed median) at $w = 25$. Shaded bands show the seed min–max range for both samplers. **(b)** Max split-$\hat{R}$ with summed NUTS divergence counts annotated: PICKLES holds $\hat{R} \approx 1.01$ with 0 divergences everywhere, while NUTS breaches the 1.01 convergence threshold at $w = 25$ ($\hat{R} = 1.13$, 3126 div over 3 seeds) and $w = 50$ ($\hat{R} = 1.08$, 343 div). All points are 3-seed medians/sums (for NUTS, each run also aggregates 4 independent vectorized-chain warmup adaptations).

1.4 Robustness on controlled ill-conditioned targets

The DES warmup sweep above shows NUTS-dense becoming *unreliable* at short warmup — erratic efficiency, non-convergence ($\hat{R}\le 1.13$), and hundreds of divergences below $w\approx 100$ — while PICKLES stays flat. On a D=21 posterior the worst NUTS cases are recoverable with a convergence check, so to isolate the mechanism and push it past the point of no recovery we use two controlled targets with exact ground truth: an ill-conditioned Gaussian ($\kappa = e^6$, d=50, randomly rotated; its dense mass matrix has 1275 free entries) and the Rosenbrock banana (d=20, position-dependent curvature no fixed metric captures). We sweep the gradient-sampler warmup from 10 to 500 and report ESS / grad, worst-coordinate bias z-score (vs. the analytic moments), and NUTS divergences, as 10-seed medians (fixed target, varying sampling seed; bands are the 10–90th percentile) (Table 6, Figure 4).

Table 6: Warmup robustness on ill-conditioned targets (10 sampling seeds on a fixed target each — the standard sampling error band). Median ESS / grad (bias z); NUTS also lists median divergences per seed and uses a max-tree-depth cap of 8. PICKLES (γ=0.5, $L_{\max}$=20, w_s=200) reports zero divergences at every warmup; its inter-run CV is 1–8% (Gaussian) / 7–30% (Rosenbrock), reflecting its run-to-run reproducibility.

	PICKLES		NUTS-dense		
warmup	ESS/grad	bias z	ESS/grad	div.	bias z
Ill-conditioned Gaussian ($\kappa = e^6$, d=50):					
10	**1.3×10^{-1}**	2.1	3.9×10^{-4}	0	1.8
25	**1.3×10^{-1}**	2.1	7.0×10^{-6}§	138,113	2.1
50	6.9×10^{-2}	2.2	1.2×10^{-3}	0	2.3
100	4.6×10^{-2}	2.3	3.7×10^{-2}	0	2.0
500	4.6×10^{-2}	1.9	**1.0×10^{-1}**	0	2.4
Rosenbrock (d=20):					
10	3.5×10^{-4}§	15.2	3.3×10^{-5}§	8,282	4.6
25	2.3×10^{-4}§	6.6	5.9×10^{-6}§	83,213	3.9
50	4.7×10^{-4}	2.7	3.4×10^{-5}§	10,271	3.1
100	6.4×10^{-4}	2.1	6.5×10^{-4}	1,083	2.2
200	5.4×10^{-4}	1.7	7.0×10^{-4}	358	2.2

§ Unreliable, greyed out: the PICKLES cells fail convergence ($\hat{R} > 1.2$, std-error > 25%), and the NUTS cells have a 6–71% divergence rate (thousands of divergences), so their ESS / grad is not a meaningful efficiency. See convergence validation below.

On the ill-conditioned Gaussian the picture is clean and decisive. NUTS-dense is *catastrophic* at short warmup: at warmup = 25 a partially-fit mass matrix triggers $\sim 1.4\times10^5$ divergences (median over 10 seeds) and ESS / grad drops to $\sim 7\times10^{-6}$ — four orders of magnitude below PICKLES. NUTS only becomes competitive once warmup ≥ 100, and at warmup = 500 it actually *overtakes* PICKLES (1.0×10^{-1} vs. 4.6×10^{-2}): a fully-converged dense mass matrix is very efficient. This is the crossover that frames the whole comparison — *NUTS-dense can be the more efficient sampler, but only after paying a long, target-dependent warmup to fit its mass matrix.* PICKLES is flat ($\sim$ 0.05–0.13) and unbiased ($z\approx 2$) at *every* warmup down to 10, because the empirical-covariance preconditioner is read off the live ensemble each iteration and there is no mass matrix to warm up.

Rosenbrock is harder and the story is more honest: its position-dependent curvature defeats any fixed metric, so *both* samplers degrade at very short warmup. But the failure modes differ sharply. PICKLES stays divergence-free and *converges* as warmup grows (bias $z=15.2\to2.7$ by warmup = 50), whereas NUTS-dense suffers 0.8–8.7×10^4 divergences through warmup = 50 and matches PICKLES only by warmup $\geq$100. Across both targets the message is the same: the affine-invariant ensemble preconditioner removes the mass-matrix warmup that makes dense-mass NUTS fragile on ill-conditioned geometry.

1.5 Convergence and fairness validation

Convergence validation (ESS is not taken on faith). ESS measures within-chain mixing and can be large even when a sampler is biased or under-dispersed, so every ESS / grad we quote is backed by independent

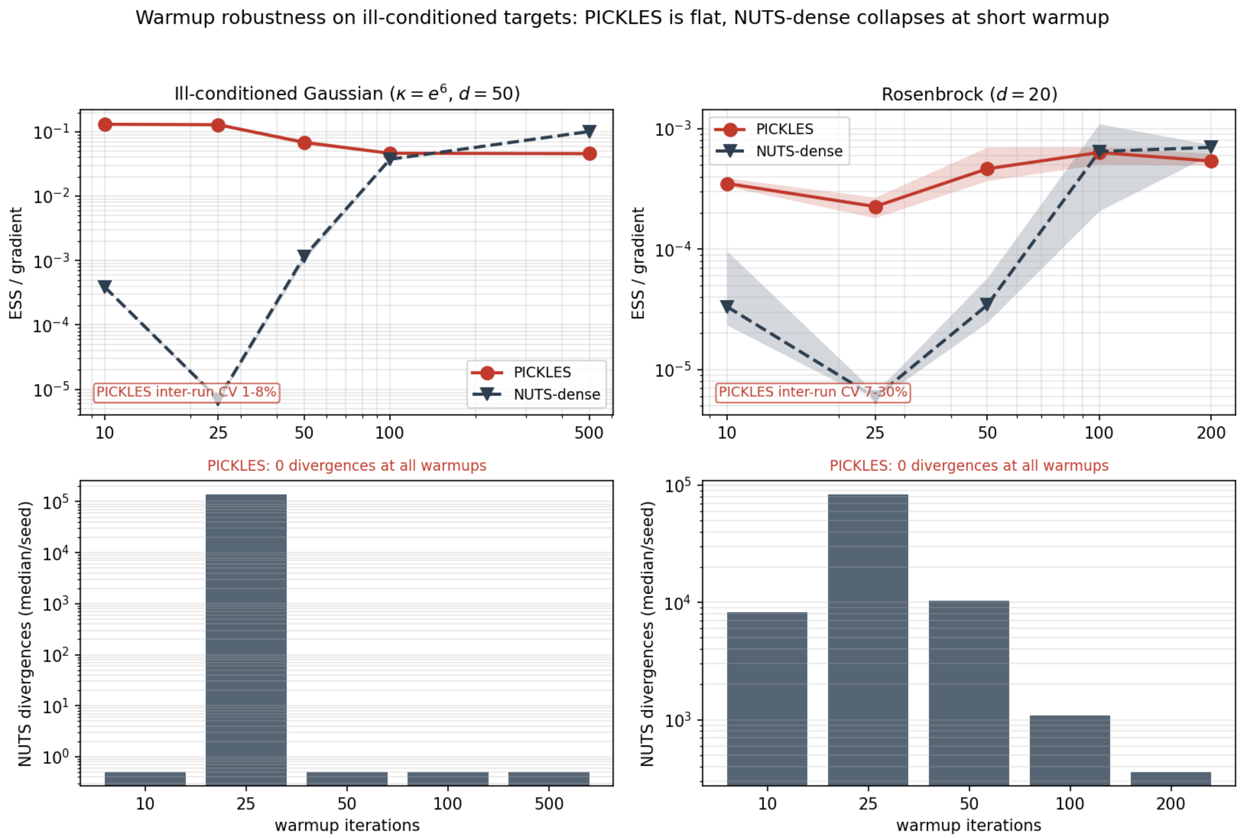

Figure 4: Warmup robustness on ill-conditioned targets. Bands are the standard sampling error — one fixed target each, varying only the sampling seed over 10 seeds (both panels). **Top:** median ESS / grad vs. warmup with 10–90th-percentile bands, PICKLES (solid) vs. NUTS-dense (dashed). PICKLES's band is small (annotated inter-run CV) because it is reproducible; NUTS's is wide at short warmup because its fitted mass is a different gamble each run. **Bottom:** NUTS divergences (median/seed); PICKLES has zero everywhere. NUTS uses a max-tree-depth cap of 8. Gaussian w=200 omitted (a `numpyro` window-adaptation schedule artifact, ESS/grad $\approx 8 \times 10^{-3}$, non-monotone vs. w=100/500).

convergence and *accuracy* checks. On the synthetic targets, where the moments are analytic, we verify rank-normalized split $\hat{R}$, the worst-coordinate posterior-mean bias, and the worst-coordinate *standard-deviation* recovery (the direct test for under-dispersion). For the ill-conditioned Gaussian (median over 10 sampling seeds at a fixed target), PICKLES is fully converged at *every* warmup — $\hat{R} = 1.001$, mean bias 0.006σ, std error $\leq 0.5\%$ — so its flat ESS / grad reflects genuinely correct, well-mixed sampling, not a misleadingly large number. The NUTS warmup$=25$ collapse is likewise *real*: $\hat{R} = 1.40$ and a 29% std error (the dense mass matrix is unfit), confirming the low ESS is a true failure. On Rosenbrock both samplers are unconverged at warmup$\leq$ 25–50 ($\hat{R} > 1.2$, std error $> 25\%$); we grey out those ESS / grad values rather than quote a meaningless efficiency. PICKLES reaches $\hat{R}\approx 1.04$ by warmup$=50$ and 1.03 by 100; NUTS reaches $\hat{R}\approx 1.00$ by warmup$=100$ but keeps diverging ($>$20,000 divergences) through warmup$=50$ regardless of $\hat{R}$.

Convergence validation on DES (no ground truth). The DES posterior has no analytic moments, so we validate by (i) rank-normalized split $\hat{R}$ and (ii) *cross-sampler agreement* between two independent algorithms. At $N = 5000$ both PICKLES and dense-mass NUTS reach $\hat{R} = 1.000$ on all 21 parameters. Transformed to the common physical parameter space, their posterior means agree to within ≤ 2.8 combined MCSE (consistent with Monte-Carlo noise across 21 parameters) and their posterior standard deviations agree to within 2.5% (ratio range 0.975–1.007). Two methods this different — ensemble ChEES-HMC and dense-mass NUTS — converging on the same posterior mean and variance is strong evidence that both are correct, and that the reported ESS values correspond to genuine effective samples.

Parametrization note (same geometry, fair comparison). Both samplers operate on the *same* unconstrained geometry. PICKLES runs on the hand-built z-space density, which reparametrizes the Uniform priors by $\theta = \mathrm{lo} + (\mathrm{hi}-\mathrm{lo})\,\sigma(z)$ (logistic) and the Normal priors by $\theta = \mu + \sigma_0 z$ (non-centered). NumPyro applies exactly the same transforms automatically (its interval bijector for the Uniforms, `LocScaleReparam` for the Normals); we verified numerically that NumPyro's transform and log-Jacobian equal PICKLES's to machine precision. The two pipelines therefore sample an identical density in identical coordinates — the comparison isolates the sampler, not the parametrization. The only difference is bookkeeping: PICKLES *stores* z-space draws while NumPyro transforms them back to physical units before storing. Because ESS is coordinate-dependent, **every ESS / grad value in this note is computed in the common physical parameter space** — PICKLES draws are mapped through $z \to \theta$ before the (rank-normalized) ESS estimate, with the identical estimator for both samplers, so all comparisons are strictly same-coordinate. (Measuring PICKLES in its native z-space instead shifts the recommended-tuning number by $\lesssim 15\%$ and changes no ranking; the only qualitative difference is that one w_s=50 seed appears collapsed in z-space but is well-mixed in physical space, so the physical-space tables are also the cleaner ones.) The synthetic warmup-robustness experiment (§1.4) is intrinsically same-coordinate: both samplers run on the identical `potential_fn`.

1.6 Summary

On the DES Y1 3×2pt benchmark, Metropolized PICKLES at the recommended $(\gamma, L_{\max}) = (0.5, 20)$ tuning *matches or slightly exceeds* dense-mass NUTS in per-gradient efficiency: an effective tie from a fiducial-ball start (Table 1) and a robust 1.44× edge from a matched, equilibrated start (Table 2, the apples-to-apples comparison), while *per typical dimension* NUTS is marginally ahead and PICKLES's worst-dimension advantage comes from a flatter ESS profile across coordinates (Table 4; all ESS in the common physical space). PICKLES is ∼24× above `emcee` at matched long production, with zero divergences. The empirical-covariance preconditioner plays the role of the dense mass matrix but needs no separate dual-averaging phase, making PICKLES robust to a 10× reduction in warmup budget that collapses NUTS-dense to a several-hundred-divergence failure. A short stretch pre-warmup ($w_s \geq 50$) is the single most important practical setting: it costs no gradients and removes the initialization sensitivity of the worst-mixing direction.

A Unadjusted PICKLES and ALDI on DES Y1

Metropolis-adjusted samplers are the standard choice in cosmology (guaranteed unbiasedness, no numerical divergence). For completeness we contrast the unadjusted gradient variants ALDI and unadjusted PICKLES (BAOAB integration without the Metropolis correction) on the same target; these methods use a projected-gradient-norm step-size adapter and are independent of the dual-averaging scheme. Bias is measured as $\max_d |\hat{\mu}_d - \mu^\star_d|/\sigma^\star_d$ against a long Metropolized PICKLES reference (the $N = 5000$, $\gamma = 0.5$ run of Table 3).

At a step size near the BAOAB stability boundary ($\eta = 0.10$), unadjusted PICKLES reaches ESS/grad = 1.15×10^{-2} — higher than Metropolized PICKLES, because each leapfrog step yields one sample rather than being consumed in an $L = 20$ trajectory — with bias $\leq 0.09\sigma_{\mathrm{ref}}$. The caveat is that the stability boundary must be respected by hand: at $\eta = 0.20$ BAOAB diverges to NaN, and ALDI (overdamped) diverges for $\eta \geq 0.05$. Metropolized PICKLES removes this risk entirely, which is why we recommend it as the default. (The unadjusted ESS / grad values in Table 7 use the original z-space estimator and are indicative; the qualitative bias–stability trade-off, not a precise ratio, is the point.)

B Per-seed data

The main-text tables report 3-seed medians; this appendix gives the underlying per-seed values. Cells in red are collapsed seeds (ESS / grad $< 10^{-3}$), where the empirical-covariance preconditioner is still globally mis-shaped before production; the bottleneck dimension for each is listed in Table 10. A short stretch pre-warmup ($w_s \geq 50$) removes every collapse.

Table 7: Unadjusted-method comparison on DES Y1, long-budget for fair asymptotic comparison; bias against the Metropolized PICKLES reference. The Metropolized rows are reproduced from §1.

	N_{ch}	N_{prod}	η	ESS	ESS / grad	bias
Metropolized (reference)						
PICKLES ($\gamma = 0.5$, $L_{\max} = 20$)	64	5,000	0.07 (adapt)	38,669	6.04×10^{-3}	0
NUTS-dense	4	5,000	0.13 (adapt)	2,314	3.60×10^{-3}	0
Unadjusted						
PICKLES ($\gamma = 0.5$, BAOAB, no adapter)	32	4,000	0.10	1,477	$\mathbf{1.15\times10^{-2}}$	0.09σ
PICKLES ($\gamma = 0.5$, BAOAB, no adapter)	32	4,000	0.05	270	2.11×10^{-3}	0.04σ
PICKLES ($\gamma = 0.5$, BAOAB, no adapter)	32	4,000	0.20	—	— (NaN)	diverged
PICKLES ($\gamma = 0.5$, BAOAB, adaptive)	32	10,000	0.065 (adapt)	588	1.84×10^{-3}	0.06σ
ALDI (overdamped, PGN-adapted)	64	20,000	0.009 (adapt)	1,543	1.20×10^{-3}	0.04σ

Table 8: Per-seed ESS / grad (acceptance) at $N_{\text{prod}} = 1000$, fid-ball init, $w_s = 0$, physical-space arviz ESS (underlies Table 1). $^\flat$ `emcee` is not converged at N=1000 ($\hat{R} = 1.8$); values indicative only.

sampler	seed 0	seed 1	seed 2	median
PICKLES (γ=0.5, $L_{\max}$=20)	5.11×10^{-3} (0.92)	7.09×10^{-3} (0.91)	1.74×10^{-4} (0.96)	5.11×10^{-3}
PICKLES (γ=0.5, $L_{\max}$=50)	4.92×10^{-3} (0.88)	4.75×10^{-3} (0.85)	5.56×10^{-3} (0.95)	4.92×10^{-3}
PICKLES (γ=0.5, $L_{\max}$=128)	4.83×10^{-3} (0.86)	5.07×10^{-3} (0.87)	5.26×10^{-3} (0.89)	5.07×10^{-3}
PICKLES (default γ=2)	3.97×10^{-3} (0.84)	3.67×10^{-3} (0.88)	2.09×10^{-4} (0.93)	3.67×10^{-3}
NUTS-dense	7.62×10^{-3} (0.80)	3.72×10^{-3} (0.80)	4.26×10^{-3} (0.80)	4.26×10^{-3}
NUTS-block	4.78×10^{-3} (0.80)	3.93×10^{-3} (0.80)	2.89×10^{-3} (0.80)	3.93×10^{-3}
NUTS-diag	1.79×10^{-3} (0.80)	2.37×10^{-3} (0.80)	2.29×10^{-3} (0.80)	2.29×10^{-3}
`emcee`/stretch$^\flat$	1.50×10^{-3} (0.28)	1.44×10^{-3} (0.27)	1.51×10^{-3} (0.27)	1.50×10^{-3}

With the corrected adapter the recommended warmup-sweep configuration ($\gamma = 0.5$, $w_s = 200$) has *no* collapsed seeds across all four warmup budgets (Table 9); collapses remain only in the no-pre-warmup ($w_s = 0$) rows, confirming that the residual seed sensitivity is an initialization effect cured by a short stretch pre-warmup, not a step-size-adaptation effect.

C Configuration summary for reproducibility

Common setup. Synthetic DES Y1 3 × 2pt mock data from the `jax-cosmo` forward model at the Campagne+ fiducial ΛCDM cosmology; data vector $\in \mathbb{R}^{2250}$, Gaussian likelihood. Posterior dimension $D = 21$ (six cosmology + two intrinsic-alignment + five galaxy bias + four shear-calibration + four photo-z). All samplers run in the unconstrained z-space (prior $\mathcal{N}(0, I_{21})$). Single NVIDIA H100 GPU, float64. ESS from the monotone-positive Sokal estimator without the $\tau \geq 1$ clamp; $\text{ESS}_{\min}$ is the worst of $D = 21$. Gradient counts are production-only. **All gradient samplers use the corrected dual-averaging step-size gain.**

Per-table settings.

- Table 1 ($N_{\text{prod}} = 1000$): 3 seeds; fid-ball init $z \sim \mathcal{N}(z_{\text{fid}}, 0.5^2 I)$; 500 warmup; $w_s = 0$. PICKLES $N_{\text{ch}} = 64$ (`vmap`), $L = L_{\max} \in \{20, 50, 128\}$, $\gamma \in \{0.5, 2\}$. `emcee` $N_{\text{ch}} = 64$, $a = 2$. NUTS $N_{\text{ch}} = 4$ sequential, max-tree-depth 7, target-accept 0.8, `init_to_median`.
- Table 2 (matched init): PICKLES prior-median init $z \sim \mathcal{N}(0, 0.5^2 I)$ with $w_s = 200$ stretch pre-warmup; 500 warmup; otherwise as above.
- Table 3: single seed, $N_{\text{prod}} = 5000$; `emcee` $N_{\text{prod}} = 50{,}000$.
- Table 4: 3 seeds; PICKLES $\gamma = 0.5$, $L_{\max} = 20$, fid-ball $w_s = 0$, $w = 500$.
- Table 5: prior-median init; PICKLES $\gamma = 0.5$, $L_{\max} = 20$, $w_s = 50$; NUTS-dense. Warmup $\in \{25, 50, 100, 200\}$; 3 seeds (one NUTS run also aggregates 4 vectorized-chain warmup adaptations).

Table 9: Per-seed ESS / grad (acceptance) for the warmup sweep (prior-median init, physical-space ESS), organized by stretch pre-warmup w_s. With $w_s \geq 50$ no seed collapses; only w_s=0 shows collapses (red, ESS / grad $<10^{-3}$).

config	seed 0	seed 1	seed 2	median
PICKLES $\gamma = 0.5$, $w_s = 200$ *(recommended):*				
warmup = 50	6.24×10^{-3} (0.92)	5.57×10^{-3} (0.94)	4.48×10^{-3} (0.94)	5.57×10^{-3}
warmup = 100	4.74×10^{-3} (0.94)	6.19×10^{-3} (0.94)	6.57×10^{-3} (0.94)	6.19×10^{-3}
warmup = 150	6.71×10^{-3} (0.94)	5.70×10^{-3} (0.91)	6.98×10^{-3} (0.91)	6.71×10^{-3}
warmup = 200	7.24×10^{-3} (0.92)	5.24×10^{-3} (0.87)	7.15×10^{-3} (0.93)	7.15×10^{-3}
PICKLES $\gamma = 0.5$, $w_s = 50$*:*				
warmup = 50	5.36×10^{-3} (0.93)	4.33×10^{-3} (0.92)	5.83×10^{-3} (0.93)	5.36×10^{-3}
warmup = 100	4.15×10^{-3} (0.92)	6.27×10^{-3} (0.95)	4.95×10^{-3} (0.91)	4.95×10^{-3}
warmup = 150	6.48×10^{-3} (0.88)	7.54×10^{-3} (0.92)	2.61×10^{-3} (0.85)	6.48×10^{-3}
warmup = 200	6.06×10^{-3} (0.90)	4.92×10^{-3} (0.90)	6.47×10^{-3} (0.90)	6.06×10^{-3}
PICKLES $\gamma = 0.5$, $w_s = 0$*:*				
warmup = 50	2.60×10^{-3} (0.98)	2.98×10^{-4} (0.95)	3.15×10^{-4} (0.90)	3.15×10^{-4}
warmup = 100	5.11×10^{-4} (0.97)	3.50×10^{-4} (0.97)	3.17×10^{-4} (0.94)	3.50×10^{-4}
warmup = 200	6.02×10^{-3} (0.91)	2.24×10^{-4} (0.95)	3.85×10^{-4} (0.96)	3.85×10^{-4}
NUTS-dense (ESS / grad; worst-dim ESS in parentheses):				
warmup = 50	1.85×10^{-4} (34)	2.27×10^{-3} (895)	1.27×10^{-3} (522)	1.27×10^{-3}
warmup = 100	3.43×10^{-3} (794)	5.67×10^{-3} (1254)	4.04×10^{-3} (1122)	4.04×10^{-3}
warmup = 200	1.31×10^{-3} (264)	2.49×10^{-3} (784)	1.93×10^{-3} (541)	1.93×10^{-3}

Table 10: Bottleneck dimension (worst-ESS, physical space) for each collapsed seed. The collapse is a globally mis-shaped preconditioner before the ensemble equilibrates: the worst-conditioned direction varies with the run (a cosmology parameter Ω_c for fiducial-ball init, a galaxy bias for prior-median init), but a short stretch pre-warmup cures it in every case.

config	seed	bottleneck dim	ESS
PICKLES γ=0.5 $L_{\max}$=20 (fid, w_s=0)	2	Ω_c	223
PICKLES γ=2 $L_{\max}$=20 (fid, w_s=0)	2	Ω_c	267
PICKLES γ=0.5 w_s=0 warm=100	0,1,2	b_4, b_1, b_3	406–653

- Table 7: single seed; unadjusted PICKLES BAOAB, fixed $\eta \in \{0.05, 0.10, 0.20\}$; fid-ball, 200-sample burn-in for bias. (Independent of the dual-averaging scheme.)